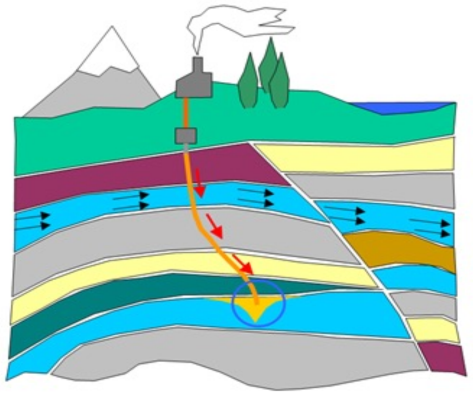
# MACHINE-LEARNED SURROGATES FOR REACTIVE TRANSPORT: NUMERICAL AND HPC INTEGRATION CHALLENGES FOR INDUSTRIAL CFD

Thibault Faney

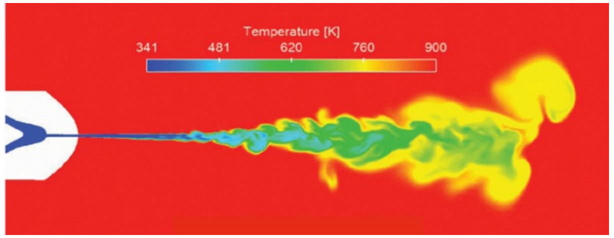
# REACTIVE TRANSPORT SIMULATION AT IFPEN

## Flow in porous media



- *reservoir / basin modeling*
- *EOR*
- *$CO_2$ sequestration*
- *$H_2$ storage*

## Combustion




- *internal combustion engines*
- *$H_2$ combustion*
- *gas turbines*

## Chemical processes

- *Refining*
- *Biofuels*
- *Batteries*

# ISSUE WITH TRADITIONAL SOLVERS

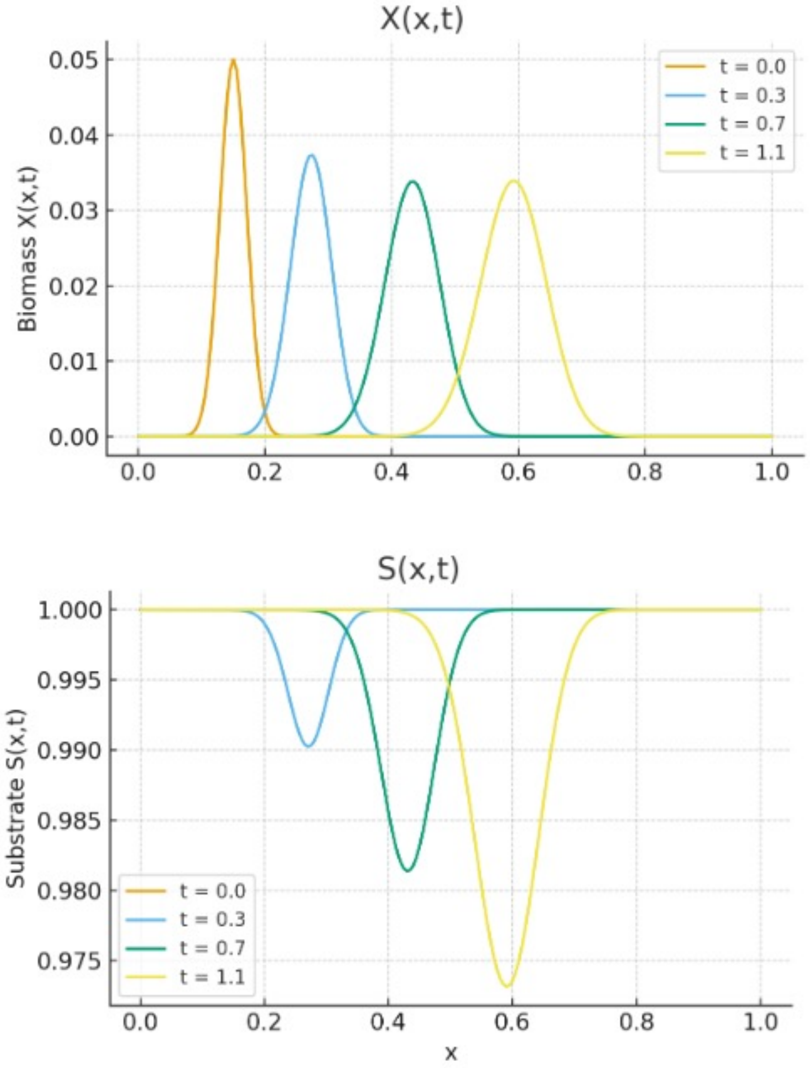
- **Reactive Transport PDE**

$$\frac{\partial \boldsymbol{u}}{\partial t} + \mathcal{T}(\boldsymbol{u}) + \mathcal{C}(\boldsymbol{u}) = 0$$

- **Monod system**

$$\frac{\partial \boldsymbol{S}}{\partial t} + vS_x = DS_{xx} - \frac{\mu S}{K+S}X$$

$$\frac{\partial \boldsymbol{X}}{\partial t} + vX_x = DX_{xx} + Y\frac{\mu S}{K+S}X$$



# ISSUE WITH TRADITIONAL SOLVERS

- **Reactive Transport PDE**

$$\frac{\partial \boldsymbol{u}}{\partial t} + \mathcal{T}(\boldsymbol{u}) + \mathcal{C}(\boldsymbol{u}) = 0$$

- **Chemistry operator $\mathcal{C}$**
  - equilibrium or kinetic reactions
  - local
  - highly non-linear

- **Solvers**
  - spatial + time discretization
  - splitting vs fully implicit methods

- **Large amount of local chemistry to be solved at every mesch cell for every time step**
  - > 75% of total simulation time

# PROMISE OF CHEMISTRY SURROGATE MODELING

- **Constraints in reactive transport**
  - Geometry
  - Initial and Boundary conditions
  - Parameters (diffusion coefficients, reaction constants, etc.)
  - Chemical system

- **Redundancy**
  - inter-simulations (optimisation, control, UQ, real-time)
  - intra-simulation (combustion front, mineral dissolution, etc.)

- **Strong potential of surrogate modeling**
  - redudancy + space parallelism !

# A PRACTICAL EXAMPLE : THERMODYNAMICAL EQUILIBRIUM

feed composition

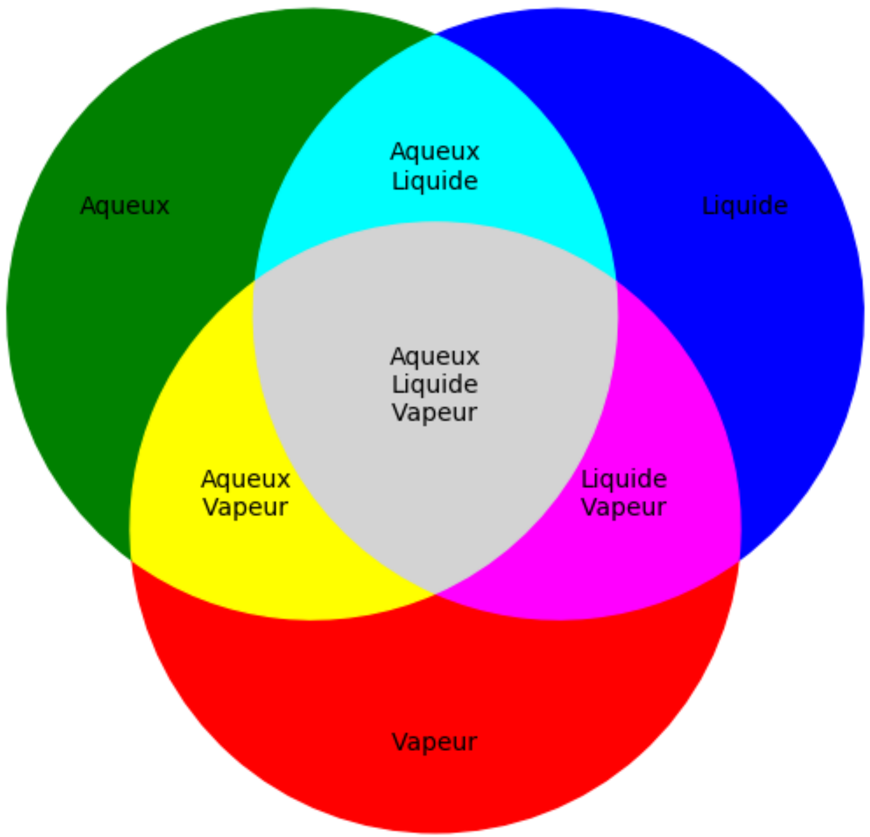

P T

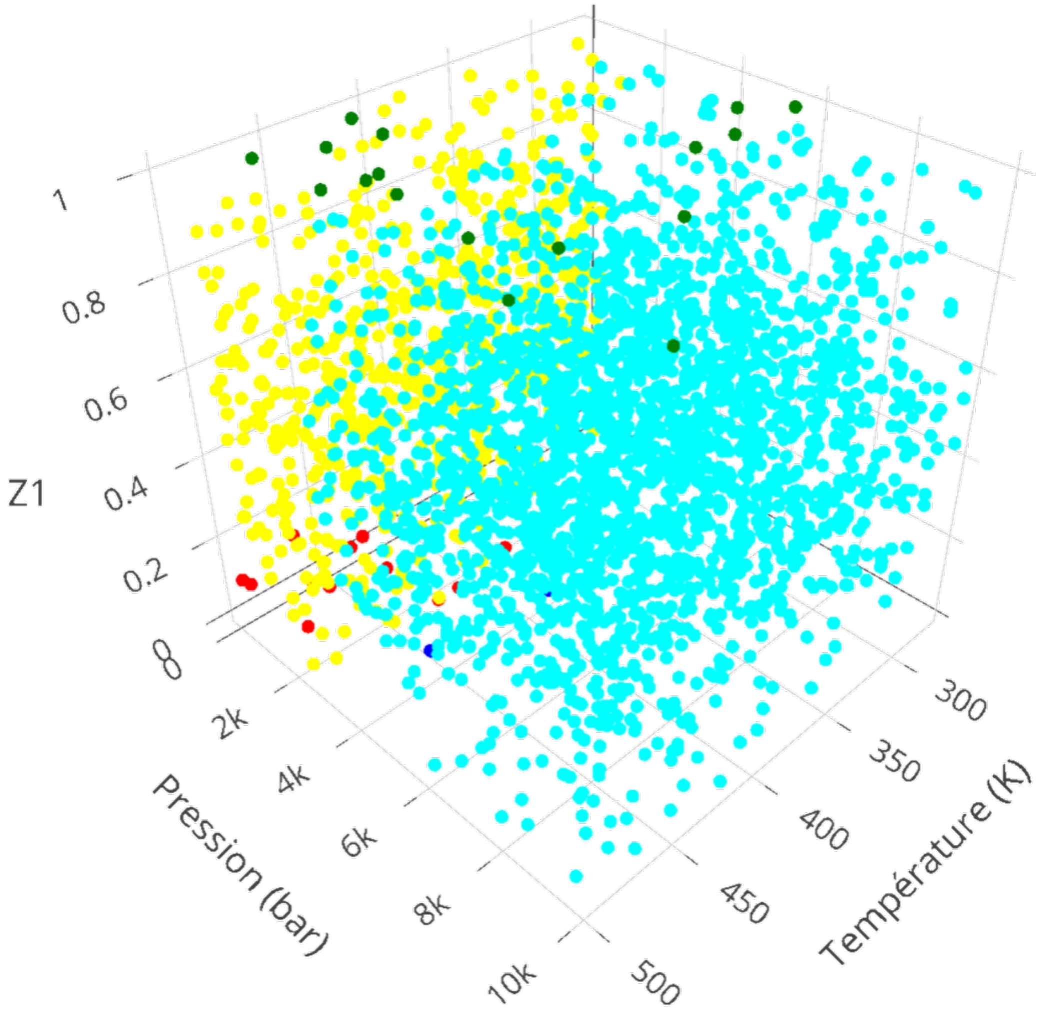
Phase Condition

+

Composition

## $H_2O$-$CH_4$ binary mixture

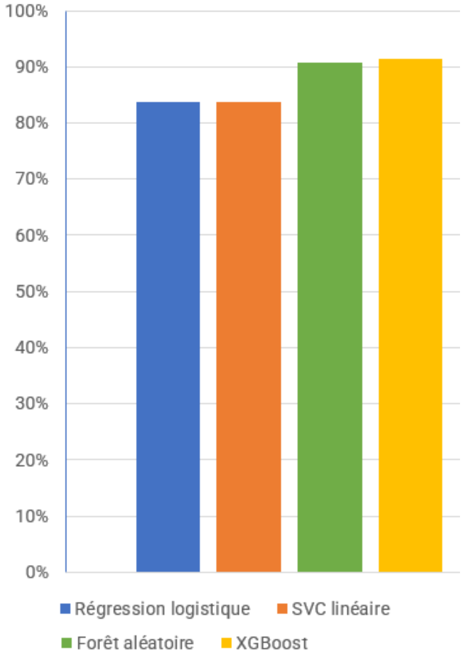


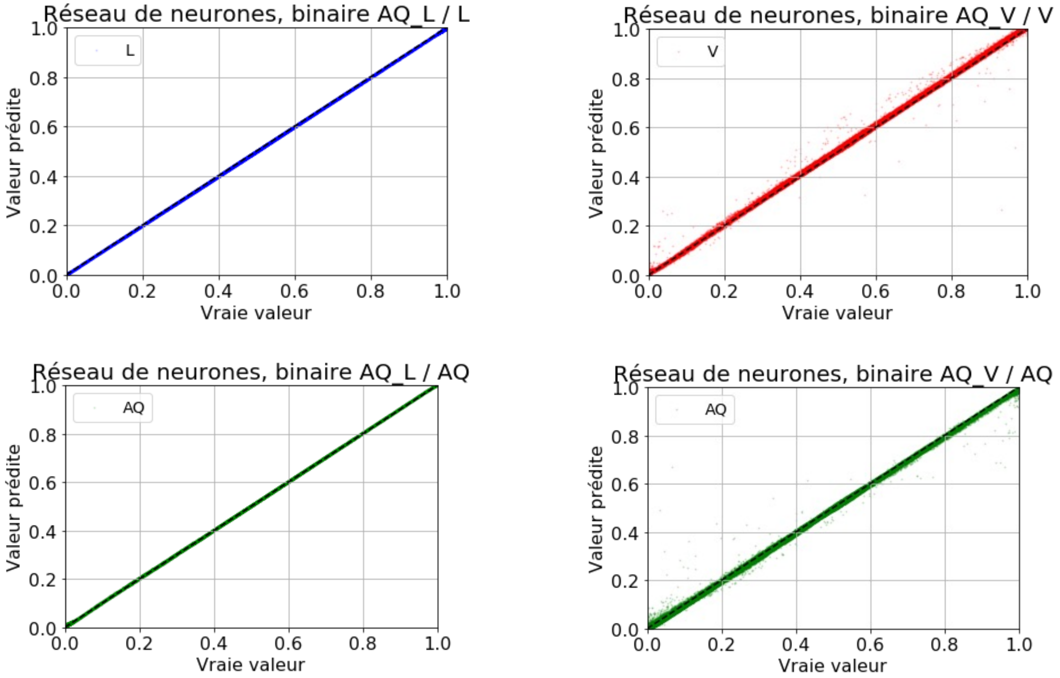
# « BLACK-BOX » SUPERVISED LEARNING MODEL

## Results

Classification Accuracy




Regression MSE = $8{,}1\times10^{-5}$




## Integration in Reactive Transport Solver (Vectorization is essential)

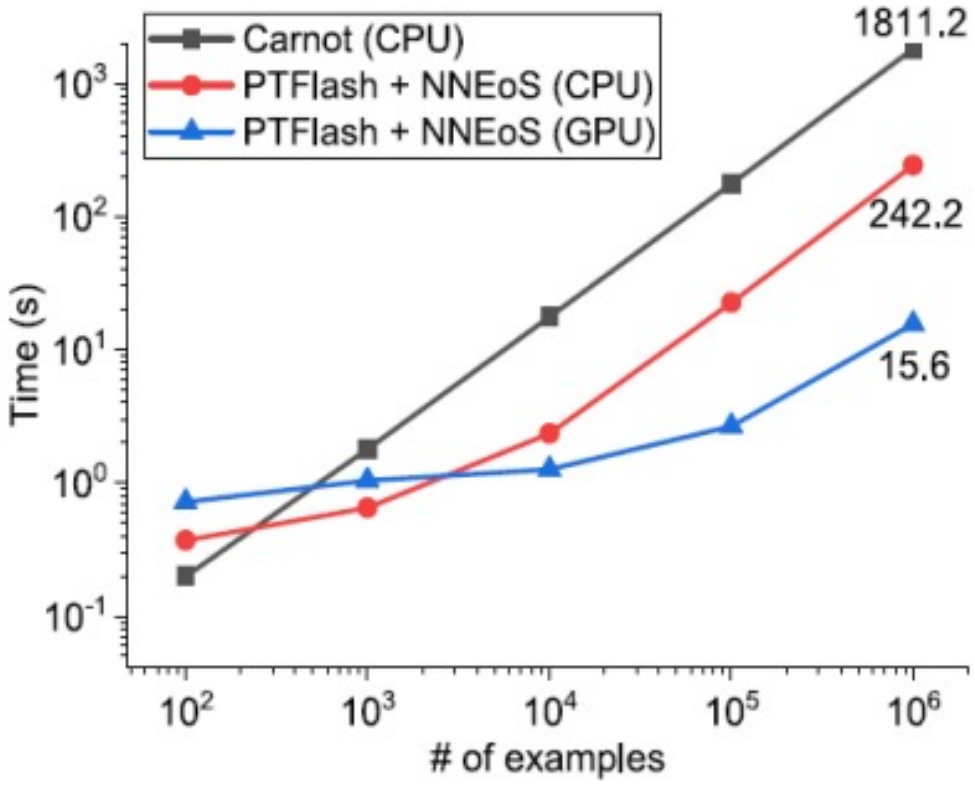
# « BLACK-BOX » SUPERVISED LEARNING MODEL

CATACLISM : **C**oupling **AI**, **T**ransport **A**nd **C**hemistry for **L**arge-scale **I**nnovative **S**imulation and **M**odeling

Local, small errors blow up in finite time

Why ? No guarantee on the solution, no physics → no stability

# « BLACK-BOX » SUPERVISED LEARNING MODEL




Performance heavily depends on parallelism

GPU not necessarily a clear winner

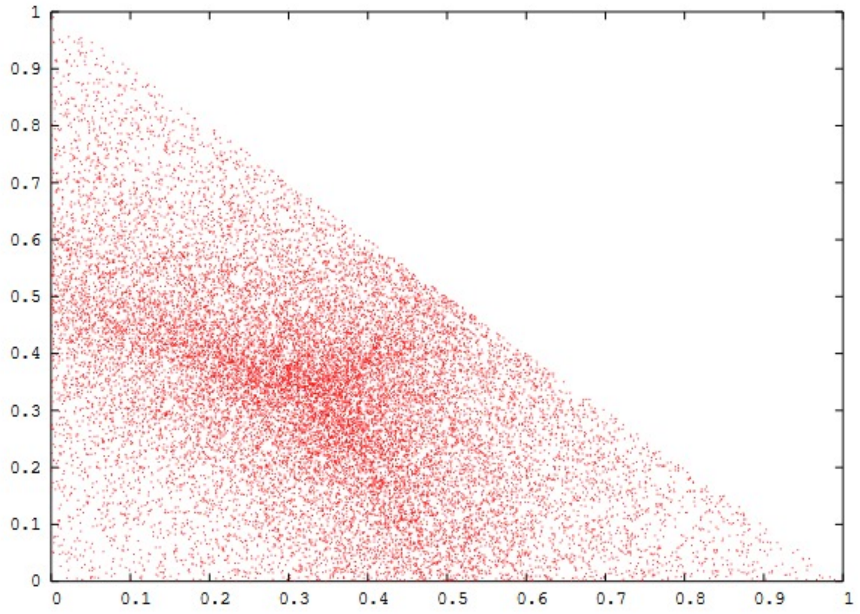
# BASIC CHALLENGES

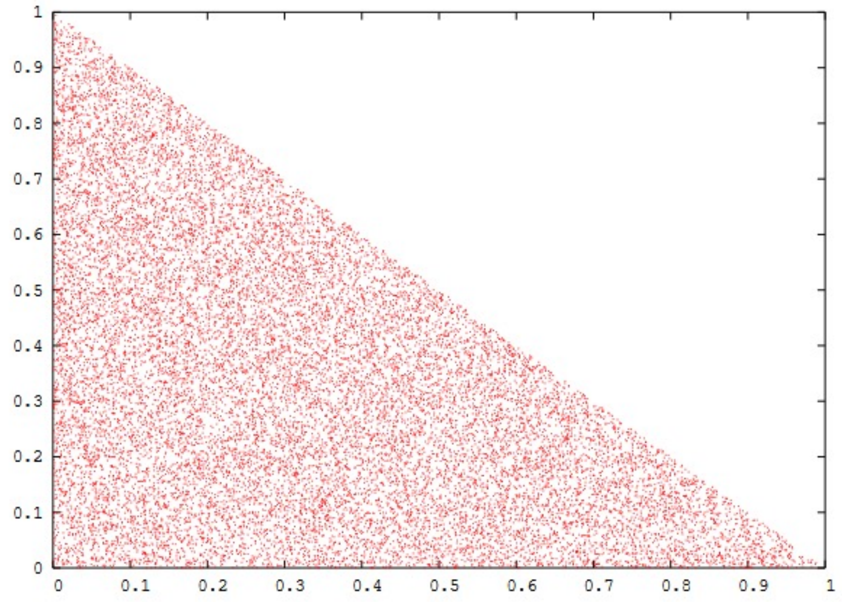
- **Numerical challenges**
  - Precision
  - Stability
  - Generalisation

- **Computational challenges**
  - Fast surrogates

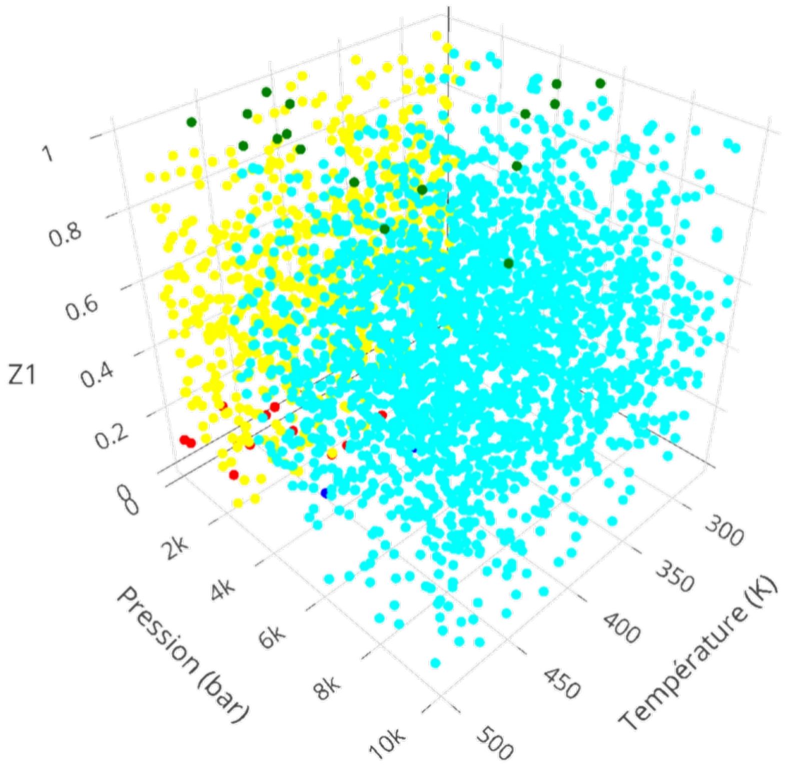
# SOME LEADS AND SOME MORE CHALLENGES

- **High dimensional sampling on simplex ($\sum z_i$=1): Dirichlet distribution**

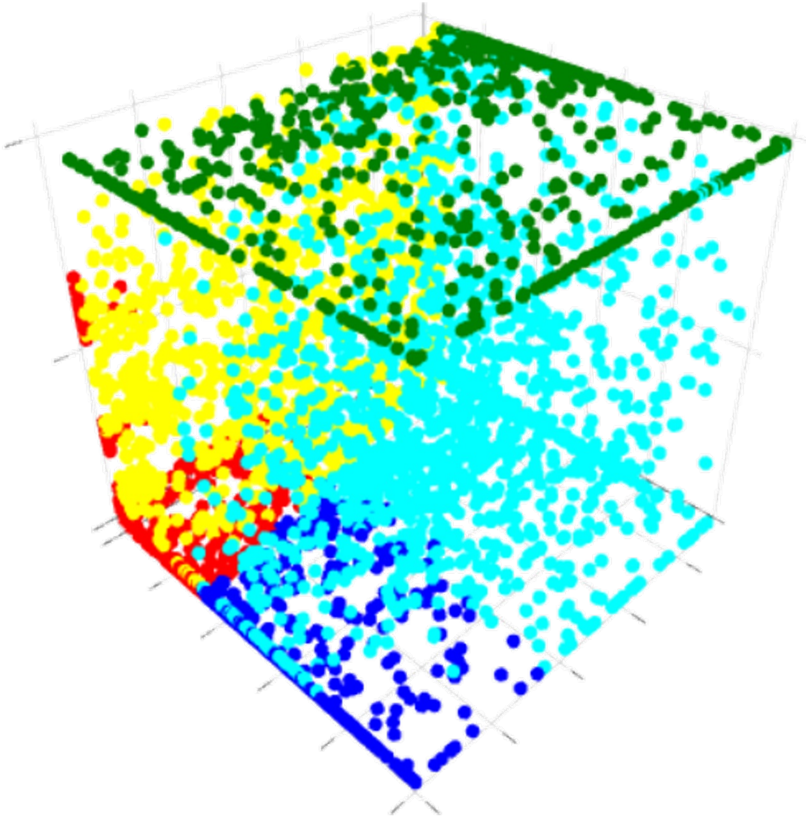


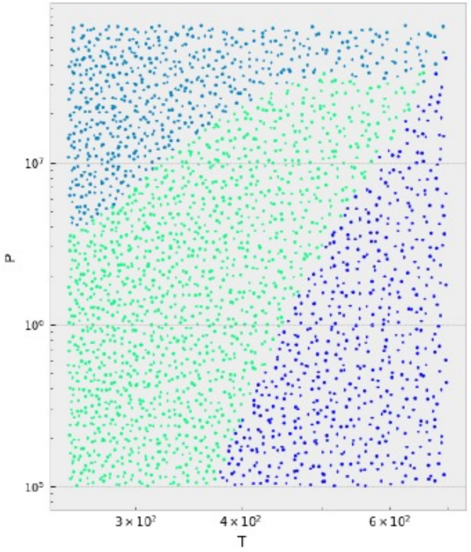
- **Unbalanced classes: Add random noise to samples**

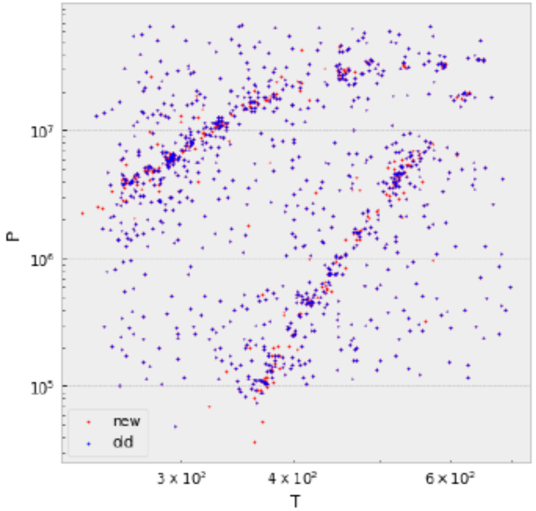



$$X_{oversampled} = X + \varepsilon$$

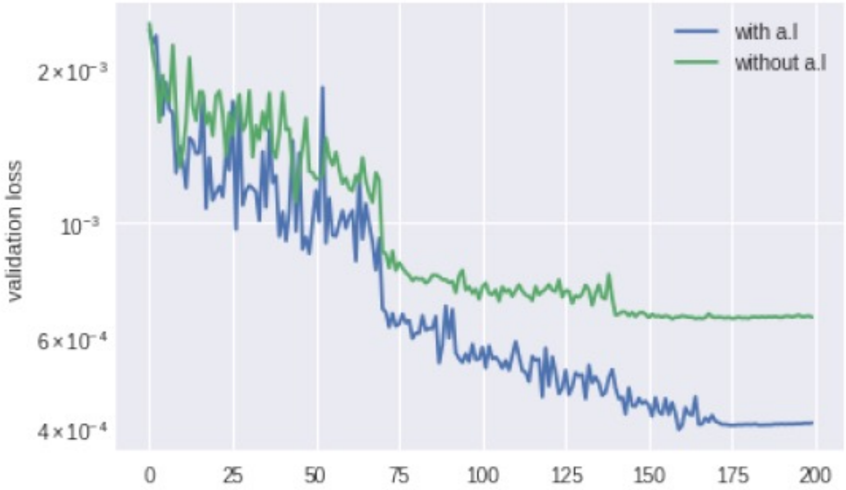


- **Active Learning**




6th step active learning



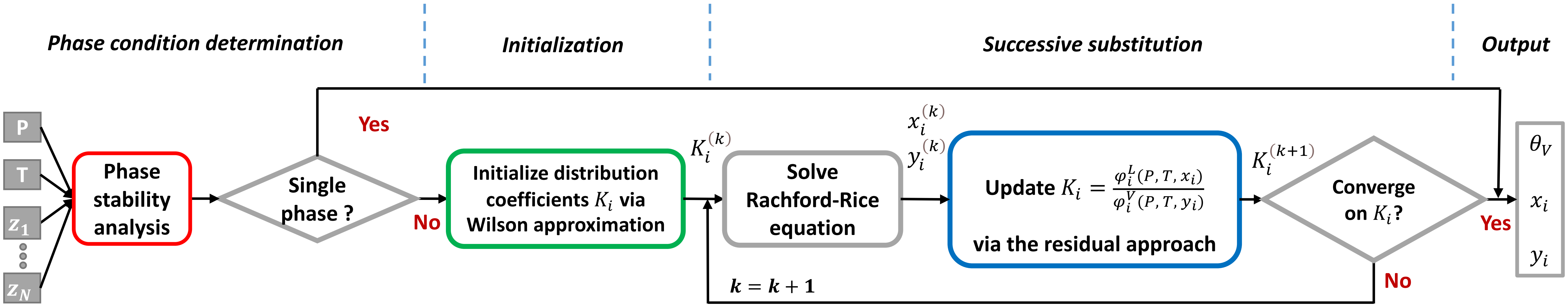
# SAMPLING

- **Online learning**
  - Flag bad predictions and add them to database
  - Retrain model periodically

- **New challenges**
  - UQ for chemistry models
  - Move data to GPU to retrain model during computation

## **Jingang Qu**'s PhD thesis (IFPEN – LIP6 at Sorbonne University)

### Numerical PT-flash calculations for vapor-liquid equilibrium




**Wilson approximation**

$$K_i = \frac{P_{c,i}}{P} \exp\left(5.373(1+\omega_i)\left(1-\frac{T_{c,i}}{T}\right)\right)$$

**Rachford-Rice equation**

$$h(\theta_V) = \sum_{i=1}^{N} \frac{(K_i - 1) z_i}{1 + (K_i - 1)\theta_V} = 0$$

$\Downarrow \theta_V$

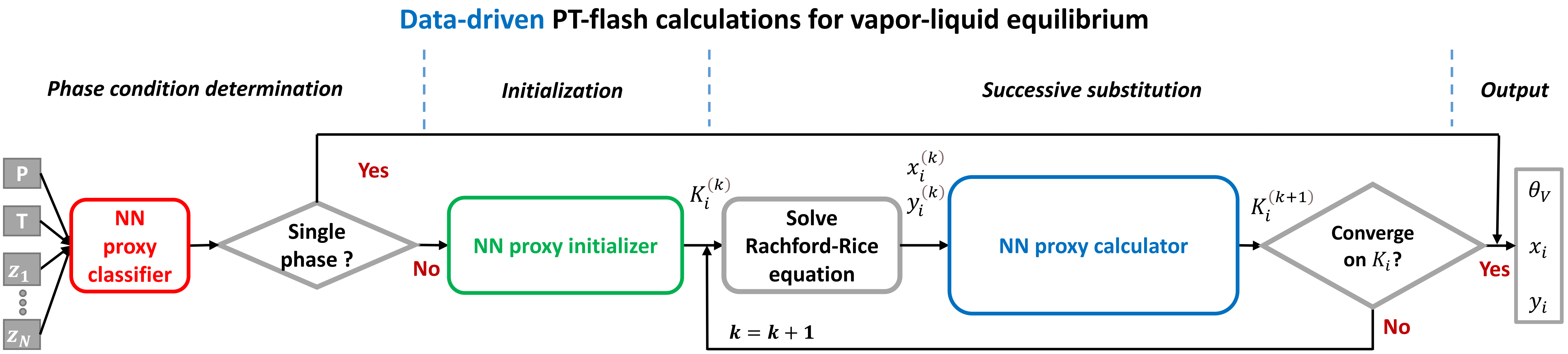
$$x_i = \frac{z_i}{1 + (K_i - 1)\theta_V}$$

$$y_i = \frac{K_i z_i}{1 + (K_i - 1)\theta_V}$$

**The residual approach**

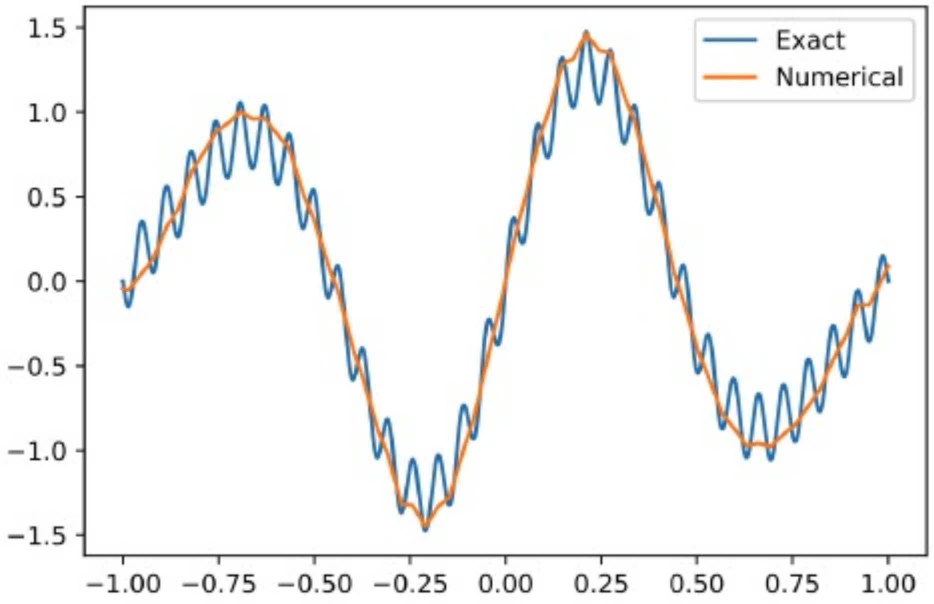
$$F = \frac{A^r(T, V, \boldsymbol{n})}{RT}$$

$$\ln \varphi_i (P, T, \boldsymbol{n}) = \left(\frac{\partial F}{\partial n_i}\right)_{T,V} - \ln Z$$


Data-driven PT-flash calculations for vapor-liquid equilibrium
Phase condition determination
Initialization
Successive substitution
Output
P
T
$z_1$
$z_N$
NN proxy classifier
Single phase ?
Yes
No
NN proxy initializer
$K_i^{(k)}$
Solve Rachford-Rice equation
$x_i^{(k)}$
$y_i^{(k)}$
NN proxy calculator
$K_i^{(k+1)}$
Converge on $K_i$?
Yes
No
$k = k + 1$
$\theta_V$
$x_i$
$y_i$

# INTRUSIVE MODELS

Neural bias of neural networks towards predicting low frequency components of the solution




Poisson equation fit with ReLU activation function[1]

Good synergy with the quadratic convergence of Newton methods **near** the solution



[1] Jinchao Xu, *CEMRACS 2023 Lecture Notes*

# INTRUSIVE MODELS

- **Convergence guaranteed**

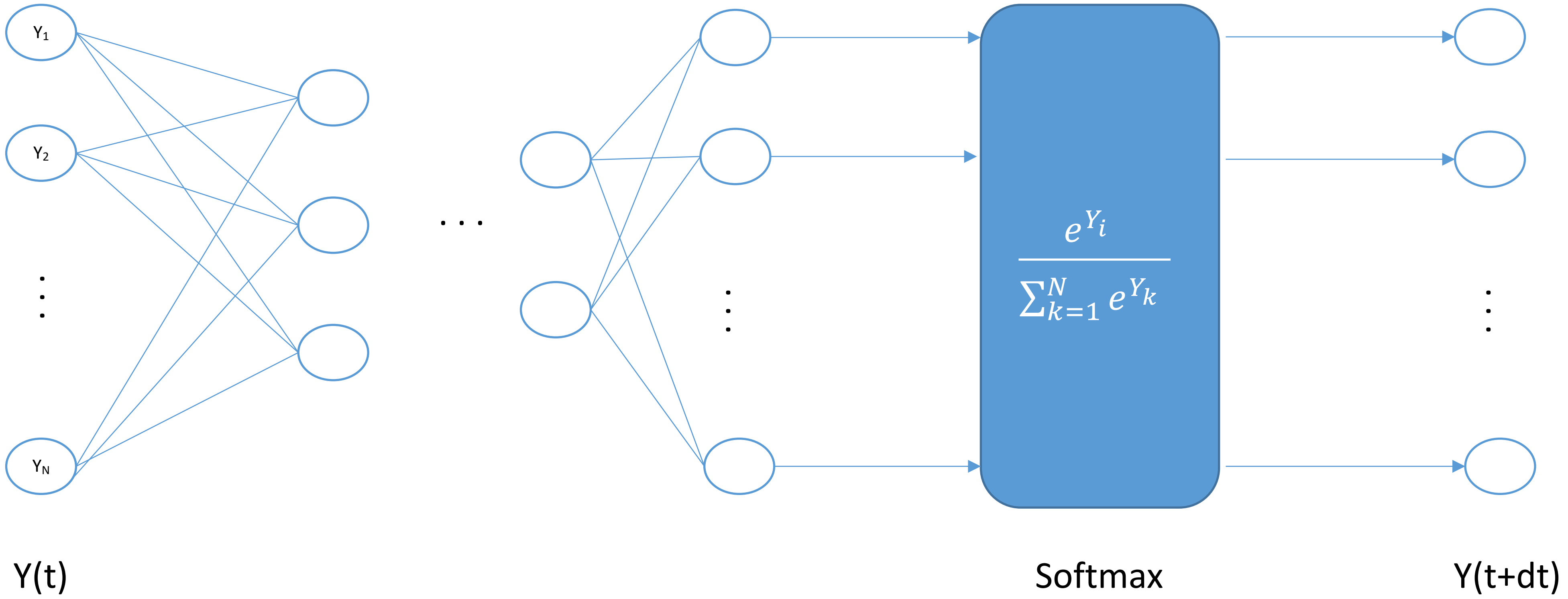
- **Downsides**
  - Very problem dependent
  - “Limited” computational gains
  - Requires low latency hybrid HPC / AI computations

# ADDING PHYSICAL CONSTRAINTS

- **Soft constraints**
  - Training issues
  - Can be difficult to express

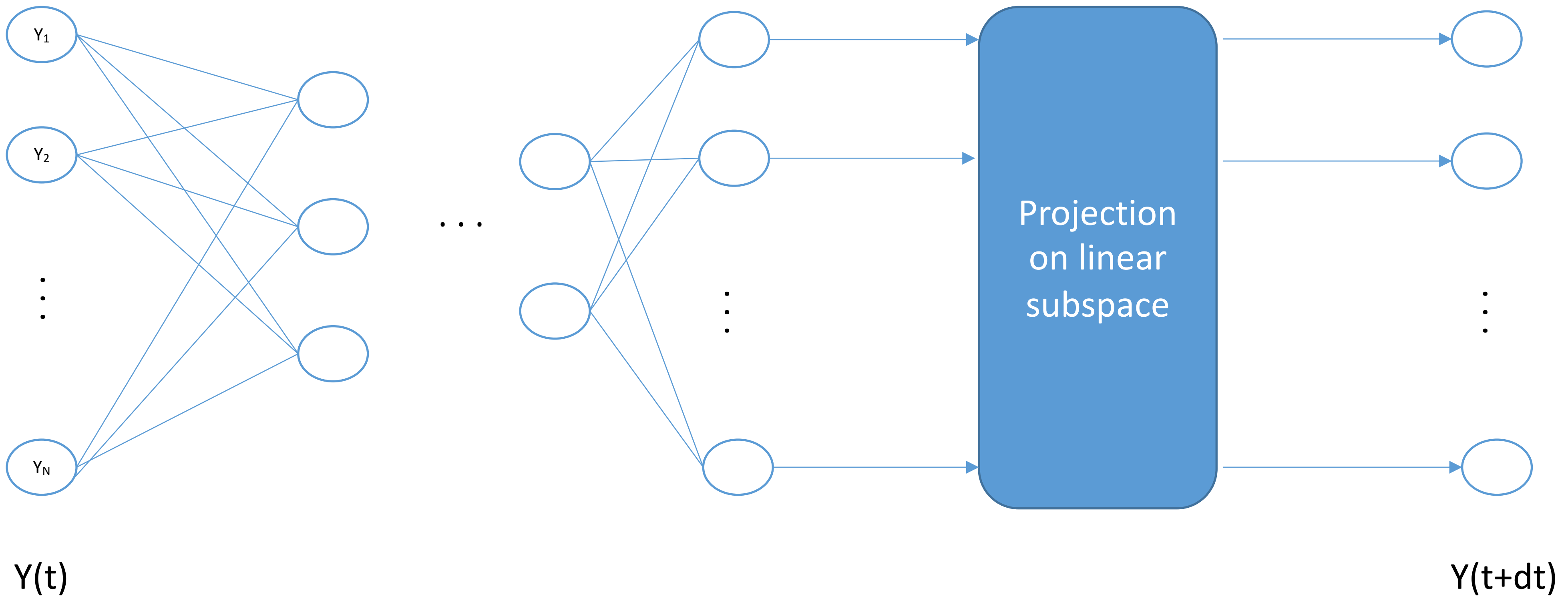
# ADDING PHYSICAL CONSTRAINTS

- **Hard constraints**
  - Problem dependent
  - Gradient issues → hard to train



# ADDING PHYSICAL CONSTRAINTS

- **Hard constraints**
  - Problem dependent
  - Gradient issues → hard to train



## Results

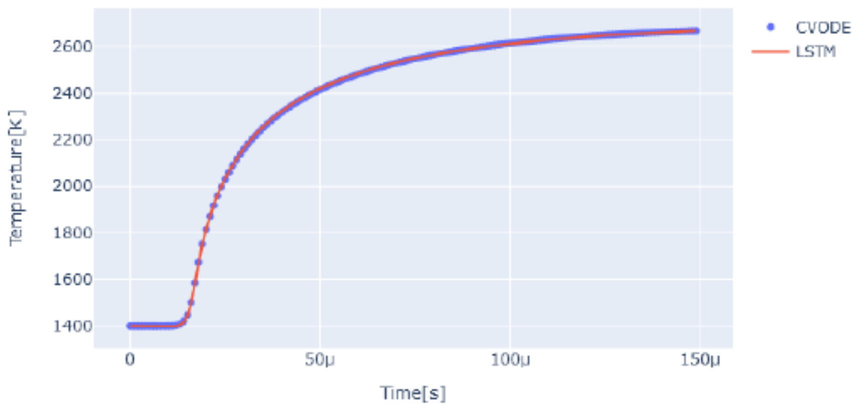



Figure 5.4: Temperature curves; $T_0$=1400 | $\phi$=0.7

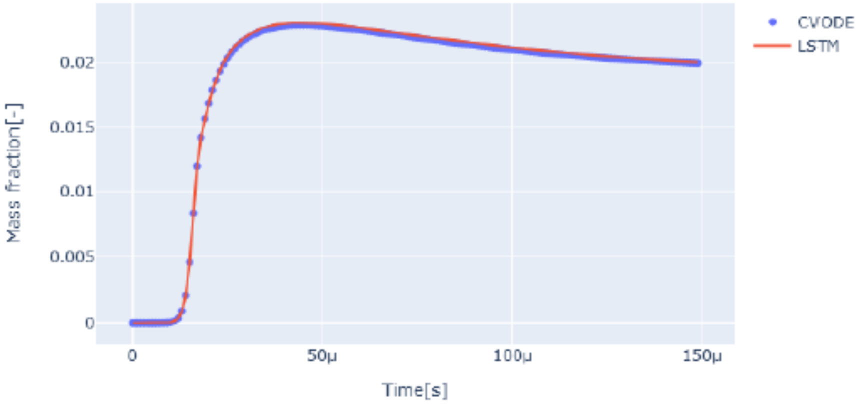
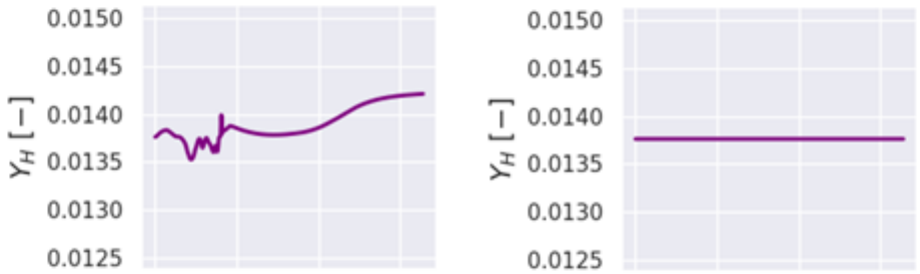



Figure 5.5: OH curves ; $T_0$=1400 | $\phi$=0.7

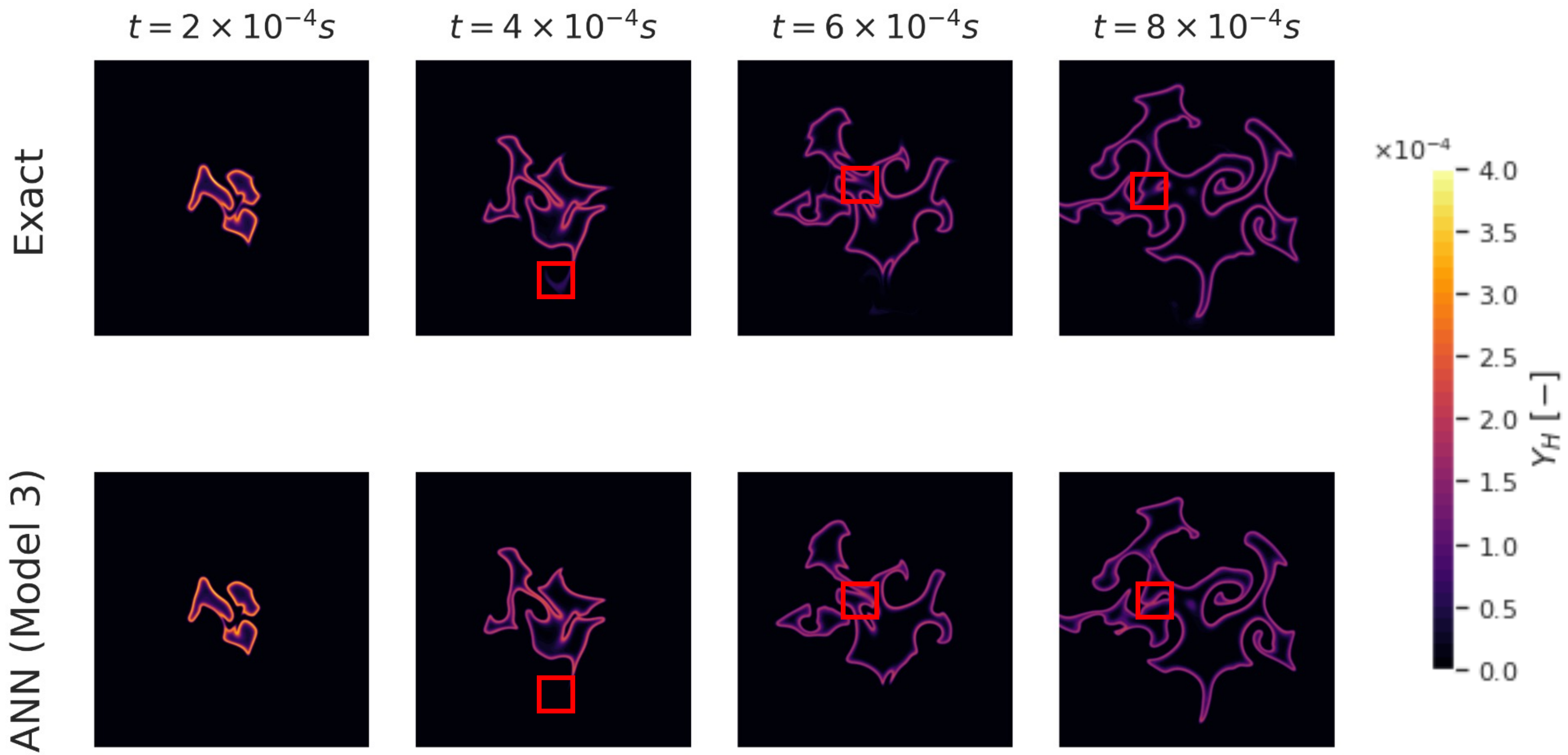



Conservation of hydrogen mass without (left) and with (right) network enforced mass conservation



Cédric Mehl, IFPEN

## Results






Cédric Mehl, IFPEN

# CHALLENGES

# CHALLENGES WITH HYBRID ML / HPC SOLVERS FOR REACTIVE TRANSPORT

- **Numerical challenges**
  - Precision and Stability
    - Intrusive models
    - Physics informed models
    - Sampling
    - UQ
    - Mixed precision integration
    - Learn over PDE trajectories
  - Generalisation
    - One model to rule several chemical systems
    - Fondational models

- **Computational challenges**
  - Online Learning
  - Low latency inference
  - Revisiting the performance / accuracy trade-off

# RECOMMENDATIONS

- **Stability**
  - A posteriori (trajectory) learning
  - Problem dependent structural form for the corrector model
  - Adding noise (both a priori and a posteriori)
  - Consistency with chemical equations (soft /hard)
  - Linearize surrogate model to study stability (eigenvalues)
- **Generalization**
  - Use probablistic classifier output (instead of hard classifier)
  - Samples need to be representative of the time evolution
  - PDE NO : use smooth rhs
  - NO with autoencoder trained on base chemical system + retrain to add new species on the fly
- **Online Learning**
  - Focus on dynamic local time stepping
  - Bayesian model to decide on the fly the use of surrogate model
  - Training a very simple domain for inactive chemical points and « track » that domain